

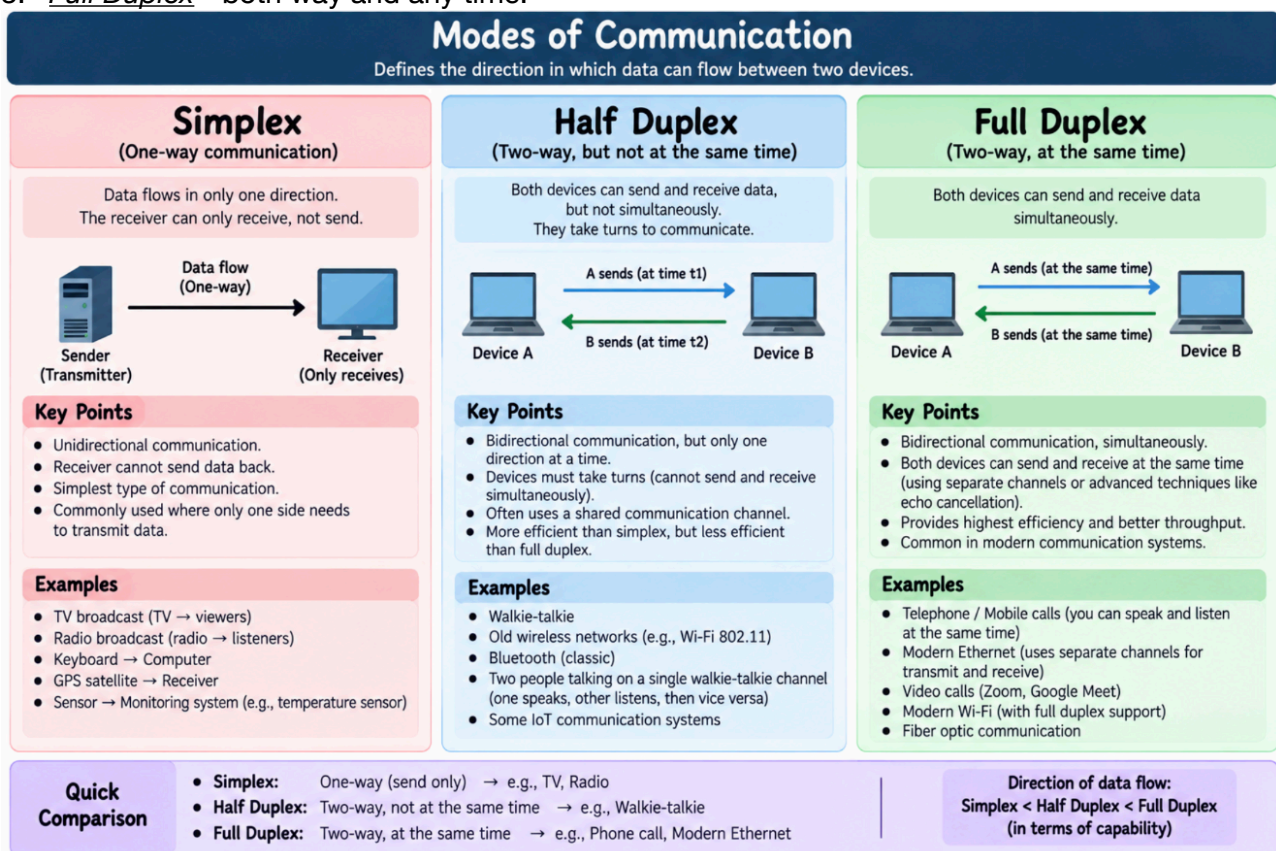
Computer Network

Terminologies:

1. Nodes - any device in the network (computers, printers, servers, routers, switches etc)
2. Protocols - set of predefined rules (TCP, UDP, HTTP, SMTP, FTP)
3. IP Address - a unique identifier assigned to each device.
4. DNS - domain name system, used to translate domain name to IP Address.
5. Peer to peer communication model - no central server (as in client server model).

Transmission Mode:

1. Simplex - one way sender and one way receiver.
2. Half Duplex - both way sender and receivers but one at a time.
3. Full Duplex - both way and any time.



Connection Types:

1. Point to Point - unique linkage between two nodes.
2. Multi Point - shared link channel between multiple nodes.

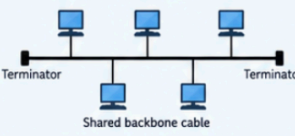
Topology:

Describes the structure of how nodes are connected to each other in a network.

1. Point to Point -
2. Mesh - each pair of nodes has connection, high cost, so highly robust,
3. Star - SPOF on central device,
4. Bus - single backbone link, data broadcasted to all nodes, SPOF, low cost and low scalability, collisions.

5. Ring - token based (so no collision), SPOF solved using dual rings, for heavy loads, costly to add more nodes.
6. Tree - hierarchical combination of star, fault isolation (if parent fails - grandparents work).

1. Bus Topology



Definition:
All nodes are connected to a single shared communication line (bus). Data is broadcast to all nodes.

Advantages:

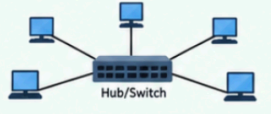
- Simple and inexpensive
- Uses less cable
- Easy to set up for small networks

Disadvantages:

- If backbone fails, entire network goes down
- Hard to troubleshoot
- Performance drops with more nodes (collisions)

Speed	Low (collisions)
Robustness	Low
Scalability	Low
Cost	Low
Use case	Small, temporary networks (legacy Ethernet)

2. Star Topology



Definition:
All nodes are connected to a central device (hub or switch). All communication goes through the central device.

Advantages:

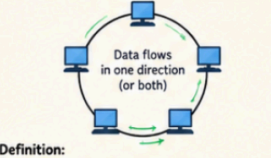
- Easy to install and manage
- If one node/link fails, others are unaffected
- Good performance with switch (no collisions)

Disadvantages:

- Central device is single point of failure
- Requires more cable
- Higher cost than bus

Speed	High (with switch)
Robustness	Medium (central device failure)
Scalability	High
Cost	Medium
Use case	Most common in modern Ethernet (LANs)

3. Ring Topology



Definition:
Each node is connected to exactly two other nodes, forming a closed loop (ring). Data passes from one node to the next.

Advantages:

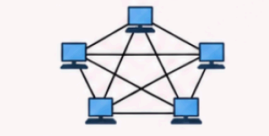
- No collisions (token based)
- Fair access for all nodes
- Performs well under heavy load

Disadvantages:

- If one node/link fails, entire ring can break (though dual ring solves this)
- Hard to add/remove nodes
- Delay increases with more nodes

Speed	Medium
Robustness	Low (single ring) High (dual ring)
Scalability	Medium
Cost	Medium
Use case	Token Ring, FDDI (legacy), some industrial networks

4. Mesh Topology



Definition:
Each node is connected to multiple (or all) other nodes. Data can take multiple paths between source and destination.

Advantages:

- Very high reliability
- Multiple redundant paths
- Fault tolerant
- Good for high traffic

Disadvantages:

- Expensive (many links)
- Complex to install and manage
- Not practical for large number of nodes (fully connected mesh)

Speed	High
Robustness	Very high
Scalability	Low (especially full mesh)
Cost	High
Use case	WANs, critical systems, military networks

5. Tree Topology (Hierarchical)



Definition:
A hierarchical combination of star topologies arranged in a tree structure (root → branches).

Advantages:

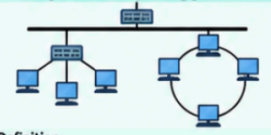
- Scalable
- Easy to manage
- Fault isolation (failure in one branch doesn't affect others)
- Widely used in organizations

Disadvantages:

- If root/backbone fails, large part of network goes down
- More cabling required
- Higher cost

Speed	High (with switch)
Robustness	Medium (root failure critical)
Scalability	High
Cost	Medium to high
Use case	Enterprise networks, campus networks

6. Hybrid Topology



Definition:
A combination of two or more different topologies (e.g., star-bus, star-ring) to meet specific requirements.

Advantages:

- Flexible and scalable
- Can be optimized for specific needs
- Combines benefits of multiple topologies

Disadvantages:

- Complex design and management
- Expensive
- Troubleshooting is harder

Speed	Varies (generally high)
Robustness	High (depends on design)
Scalability	High
Cost	High
Use case	Large and complex networks (e.g., enterprise, data centers)

7. Point-to-Point Topology



Definition:
A direct link between two nodes. Data flows only between these two nodes.

Advantages:

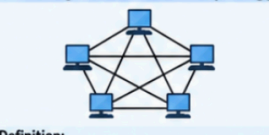
- Simple
- High speed
- Secure (dedicated link)
- No interference

Disadvantages:

- Only connects two nodes
- Not scalable
- Not cost-effective for larger networks

Speed	Very high
Robustness	High (dedicated link)
Scalability	Low
Cost	Low (for two nodes)
Use case	WAN links, leased lines, router-to-router links

8. Fully Connected Topology



Definition:
A special case of mesh topology where every node is connected to every other node.
Number of links = $\frac{n(n-1)}{2}$ for n nodes.

Advantages:

- Maximum reliability
- Multiple redundant paths
- Excellent performance
- No single point of failure

Disadvantages:

- Very high cost
- Uses a large number of links/ports
- Not scalable (becomes impractical for large n)

Speed	Very high
Robustness	Very high
Scalability	Very low
Cost	Very high
Use case	Small critical networks (e.g., military, backbone systems)

Key Takeaways:

- Topology choice depends on cost, reliability, scalability, and performance requirements.
- Star and tree are most common in modern networks.

- Mesh/fully connected give highest reliability but are expensive.
- There is no single best topology – it depends on the use case.

Types of Networks:

1. PAN - Personal, Extreme high speed, private, 100m
2. WAN - Wide, low speed, public/private, more than 50km
3. MAN - Metropolitan, 10-50km, city
4. LAN - Local, 5km, private, university

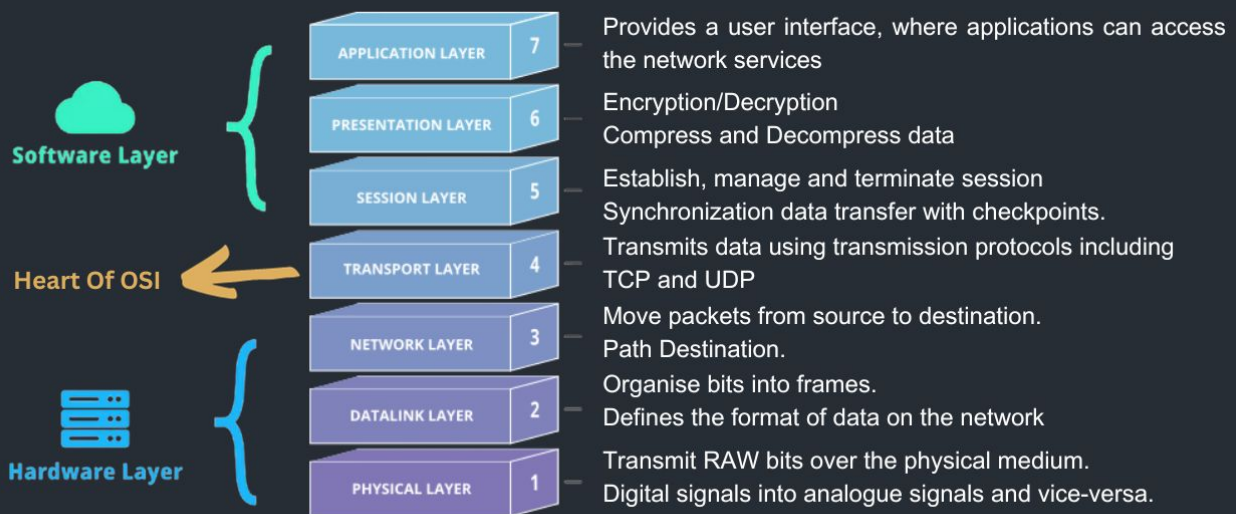
OSI Model:

OSI (Open Systems Interconnection) model is a 7-layer conceptual framework that standardises how data is communicated between devices over a network.

Sender -> data goes from App to Physical Layer.

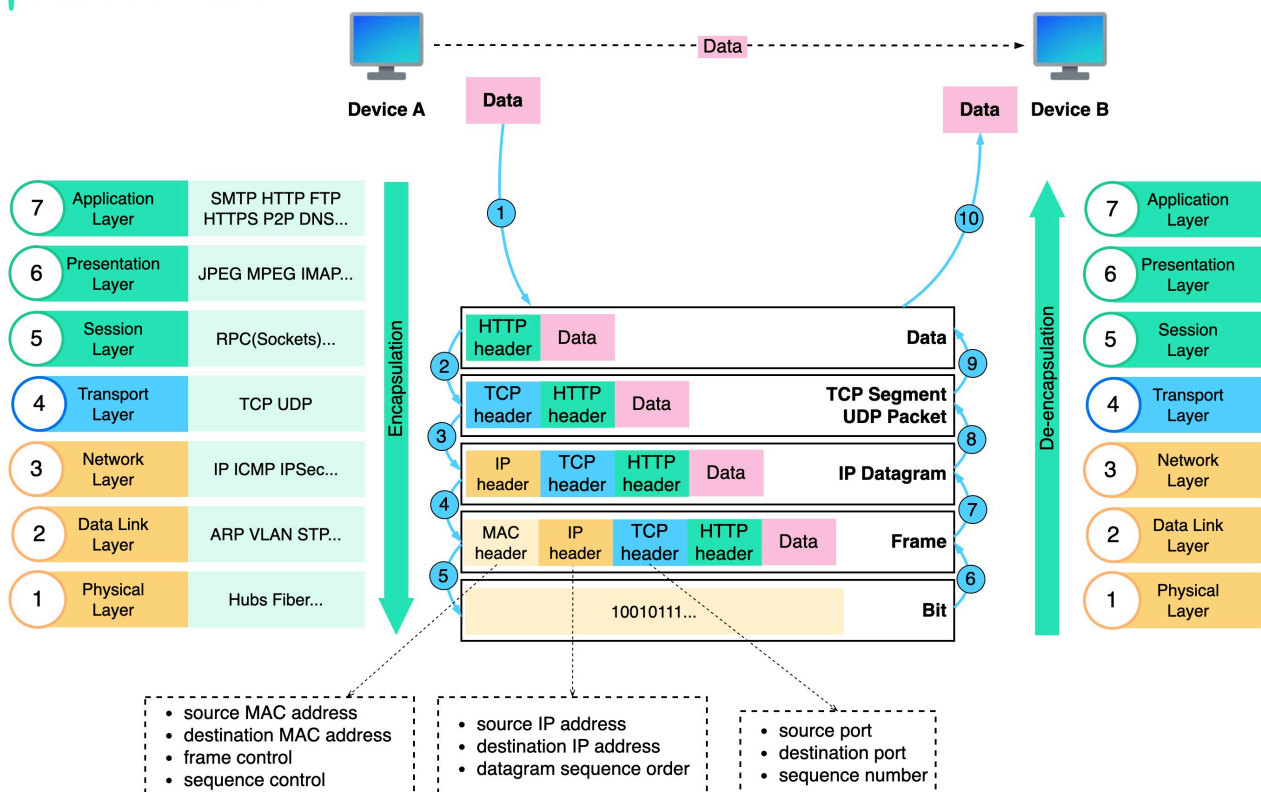
Receiver -> data goes from Physical to App Layer.

7 Layers of the OSI Model



What is OSI model

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Physical Layer:

Transmission of data (stream of 0&1 bits).

topology, transmission mode, encode and decode data stream, data transfer rate.

Types of Medium:

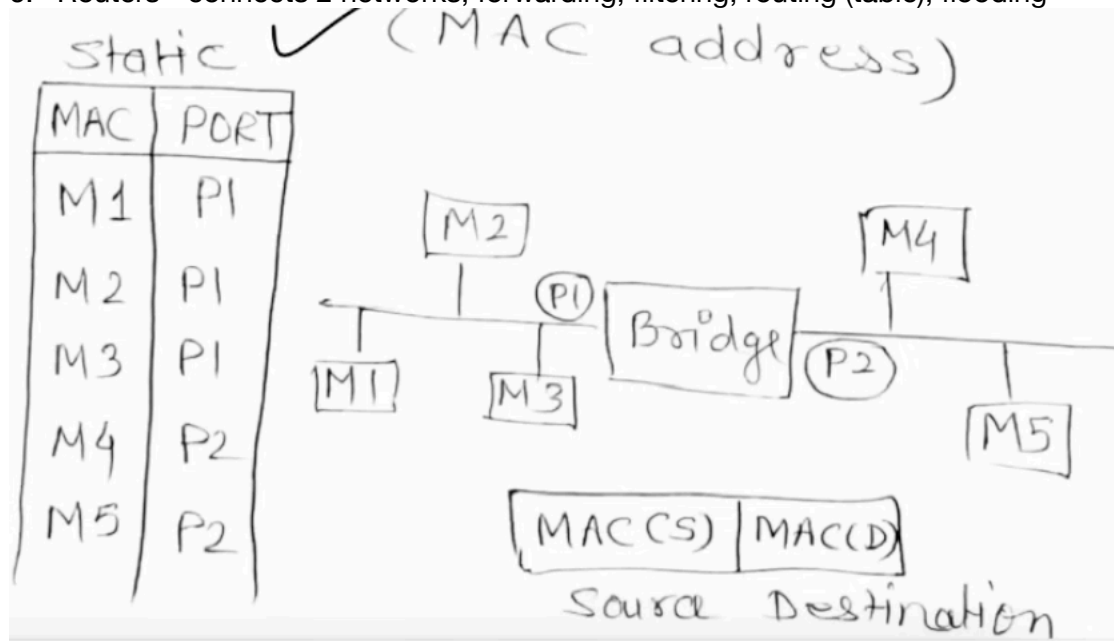
1. Guided - wired cables, twisted pair, coaxial, optical fibre (full duplex).
2. Unguided - radio, micro, infra red waves,

Types of Encoding:

1. Manchester Encoding.
2. D Manchester Encoding.

Devices in CN:

1. Cables -
2. Repeaters - 2 port hardware, no filtering only forwarding (collision occurs), sets the strength to original.
3. Hubs - multiport repeater hardware, no filtering only forwarding (collision occurs), broadcasting.
4. Bridges - s/w + h/w (store and forward), connects 2 LANs, this filters using static table but still collision on same port.
5. Switches - Multiport bridge, full duplex links SO 0 collision,
6. Routers - connects 2 networks, forwarding, filtering, routing (table), flooding



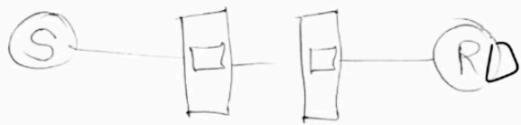
Switching:

Transferring of data packets from one device to other in network,

1. Packet - data divided into packets, 2 types - virtual and datagram,
2. Message - forwarded across network only, hop by hop delivery, high traffic, high delay.
3. Circuit - physical layer, connection setup - transmission + propagation times - tear down, in-order transmission, resources get reserved, no store and forward, low delay.

transmission time - time to keep data on link, proportional to Length/Bandwidth.

• Virtual Circuit ✓
Yes



1) BW, CPU, Buffer
Reservation

2) Path is same
(for all packet)

3) Inorder

4) Connection
oriented

No Datagram ✓



1) NO Reservation

2) May or May Not

3) out of order

4) Connection less

Data Link Layer:

gets a payload from network layer when in sender side -> attaches header and trailer to it -> forwards to PL.

Feature	LLC (Logical Link Control) (Higher)	MAC (Media Access Control) (Lower Layer)
Main purpose	Provides logical link control between Network layer and MAC	Controls access to the physical medium
Concerned with	Which network-layer protocol is being carried and logical communication	Who can transmit and when
Addressing	Generally not responsible for physical addressing	Uses MAC addresses
Device involved	Logical communication between network protocols	NICs / network interfaces
Medium access	✗ No	✓ Yes
Frame handling	Helps identify/multiplex protocols	Handles MAC framing and access control

Access Control:

Feature	Pure ALOHA	Slotted ALOHA
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Transmission time	Any time	Only at the beginning of a time slot
Synchronisation	 Not required	 Required
Collision window	2T	T
Vulnerability period	2 × frame transmission time	1 × frame transmission time
Maximum throughput	18.4%	36.8%
Collision probability	Higher	Lower
Complexity	Simpler	Slightly more complex
Main requirement	No global clock	All nodes must be synchronised
Retransmission	Random backoff	Random backoff in slots
Best suited for	Simple, loosely synchronised networks	Networks where synchronisation is possible
Feature	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access / Collision Detection	Carrier Sense Multiple Access / Collision Avoidance
Mainly used in	Traditional wired Ethernet	Wi-Fi / wireless LANs
Basic idea	Detect collision after transmission starts	Try to avoid collision before transmission
Can detect collision while transmitting?	 Yes	 Generally no
Why?	Sender can monitor the wired medium	Wireless node can't reliably listen while transmitting
If collision occurs	Stop transmission + backoff + retransmit	Use backoff/ACK mechanisms to reduce collision probability
ACK required?	 Not normally	 Typically used
Hidden terminal problem	 Not a major issue	 Major issue
RTS/CTS	 No	 Can be used
Modern relevance	Mostly historical/shared half-duplex Ethernet	Still fundamental to Wi-Fi

Flow Control:

Sliding Window Protocol techniques:

Feature	Stop-and-Wait	Go-Back-N (GBN)	Selective Repeat (SR)
Window size	1	> 1	> 1
Frames in transit	1	Multiple	Multiple

ACK type	ACK for each frame	Usually cumulative ACK	Individual/selective ACK
Out-of-order frames	Not applicable	Discarded	Buffered
Error handling	Retransmit the lost frame	Retransmit lost frame + all subsequent frames	Retransmit only lost/damaged frame
Retransmission efficiency	Low	Medium	High
Receiver complexity	Very low	Low	High
Sender complexity	Low	Medium	High
Buffer requirement	Very low	Low	High
Bandwidth utilisation	Low , especially with high RTT	Better	Best
Suitable for	Simple/slow links	Relatively reliable links	Noisy/high-delay links
Main disadvantage	Wastes bandwidth	Unnecessary retransmissions	Complex buffering/ACK management

Types of Acknowledgement:

Feature	Piggybacking	Pure Acknowledgement
Meaning	ACK is attached to an outgoing data frame	ACK is sent as a separate frame
Data + ACK	Same frame	Separate frames
Number of frames	Fewer	More
Bandwidth	Efficient	Less efficient
Overhead	Lower	Higher
Used when	Receiver also has data to send	Receiver has no data to send
Delay	ACK may be slightly delayed waiting for outgoing data	ACK can be sent immediately
Example	B receives A's data → B sends its data with ACK for A	B receives A's data → B sends ACK only

Error Control:

Method	Purpose	How it works	Error Recovery?
Parity Check	Detect errors	Add a parity bit (even/odd)	×
Checksum	Detect errors	Calculate checksum and compare at receiver	×
CRC	Detect errors	Polynomial division generates a CRC remainder	×

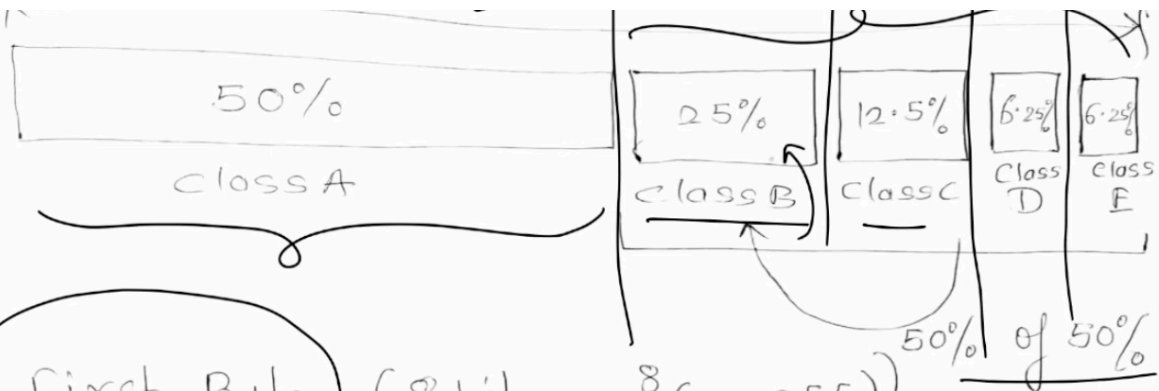
ARQ	Detect + recover	Receiver sends ACK/NAK; sender retransmits	✓
Stop-and-Wait ARQ	Reliable transfer	Send one frame → wait for ACK	✓
Go-Back-N ARQ	Reliable transfer	Retransmit erroneous frame and subsequent frames	✓
Selective Repeat ARQ	Reliable transfer	Retransmit only erroneous/lost frames	✓
FEC (Forward Error Correction)	Detect + correct	Add redundant bits so receiver can correct errors itself	✓

Network Layer:

Address	Layer	Identifies	Example
Physical / MAC Address	Data Link (L2)	Network interface/ device on local network	00:1A:2B:3C:4D:5E
Logical / IP Address	Network (L3)	Device/interface across networks	192.168.1.10
Port Address	Transport (L4)	Specific application/ process	:443
Application-specific Address	Application (L7)	Specific resource/ user/service	www.example.com, email address

Feature	IP Address	MAC Address
Full form	Internet Protocol Address	Media Access Control Address
Layer	Network Layer (L3)	Data Link Layer (L2)
Purpose	Identifies a device/interface across networks	Identifies a network interface on the local network
Type	Logical address	Physical / hardware address
Assigned by	Network configuration / DHCP / ISP, depending on context	Usually assigned to the network interface by manufacturer
Changes?	Can change	Usually persistent, though it can be changed/spoofed
Size	IPv4: 32-bit ; IPv6: 128-bit	Typically 48-bit
Used by	Routers for routing	Switches for local delivery
Scope	Local or global, depending on address	Primarily local-link
Example	192.168.1.10	00:1A:2B:3C:4D:5E

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Address space	~ 4.3 billion addresses	~ 3.4 × 10³⁸ addresses
Notation	Decimal, separated by .	Hexadecimal, separated by :
Example	192.168.1.10	2001:db8::10
Header size	20–60 bytes	40 bytes fixed
Address configuration	Manual / DHCP	SLAAC , DHCPv6, manual
Broadcast	 Supported	 No broadcast
Multicast	 Supported	 Supported
Any cast	Limited/less common	 Built-in
NAT	Widely used	Generally not required for address exhaustion
Fragmentation	Routers and hosts can fragment	Only source host fragments
Header checksum	 Present	 Removed
Address resolution	ARP	Neighbour Discovery Protocol (NDP)
Security	IPsec support	IPsec support designed into IPv6 specifications
QoS	ToS / DSCP	Traffic Class + Flow Label
Main motivation	Original Internet addressing	Huge address space + modernised networking



first Byte (8 bit $\rightarrow 2^8 (0 - 255)$)

Class A $\rightarrow 0$ to 127 [50% of 255]

Class B $\rightarrow 128$ to 191 [50% of 128]

Class C $\rightarrow 192$ to 223 [50% of 64]

Class D $\rightarrow 224$ to 239 [50% of 32]

Class E $\rightarrow 240$ to 255 [50% of 32]

IPv6

